

EE 210L — Digital Circuits and Logic Design Lab

Lab 3: Multiplexers and Demultiplexers

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record measured voltages, not expected ones. Submit this report on Canvas — see Canvas for the deadline and grading rubric.

Part 1 — Basic Multiplexer Operation

Data inputs wired: D0, D2, D4, D6 = +5 V D1, D3, D5, D7 = GND

S2	S1	S0	Selected input	V at Y (V)	Y	V at W (V)	W
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Is W always the complement of Y? _____

Part 2 — The Enable Input

Select code used: _____ Y with $\bar{E} = 0$: _____ V Y with $\bar{E} = 1$:
_____ V

What does the enable input do, and why would a shared data bus need one?

Part 3 — Three-Variable Function on the MUX

$$F(A,B,C) = A'BC + AB'C' + AB$$

Fill in which data inputs you tied HIGH, then record measurements.

A (S2)	B (S1)	C (S0)	D _n tied to	Measured V _{out} (V)	Logic Output	Expected F
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

How many logic gates did this function require? _____

Part 4 — Four-Variable Function on the MUX

$G(A,B,C,D) = \Sigma m(0, 2, 5, 7, 8, 10, 13, 15)$

4.1 Residue Table

S2 S1 S0 (A B C)	Rows covered (minterms)	F when D = 0	F when D = 1	Wire D _n to (0, 1, D, or D')
0 0 0				
0 0 1				
0 1 0				
0 1 1				
1 0 0				
1 0 1				
1 1 0				
1 1 1				

4.2 Measured Results

A	B	C	D	Measured V_{out} (V)	Logic Output	Expected G
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

Part 5 — Decoder as a Demultiplexer

Record the voltage on every output for each case. Remember the 74LS138 outputs are active LOW.

5.1 Address = 010

Data in (pin 6)	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
0								
1								

5.2 Address = 101

Data in (pin 6)	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
0								
1								

Which output followed the data input in each case, and what did the other seven do?

Discussion and Conclusions

Address at least: why implementing a function on a MUX needs no minimization, what the residue method buys you, and how a decoder becomes a demultiplexer.

[illegible]

